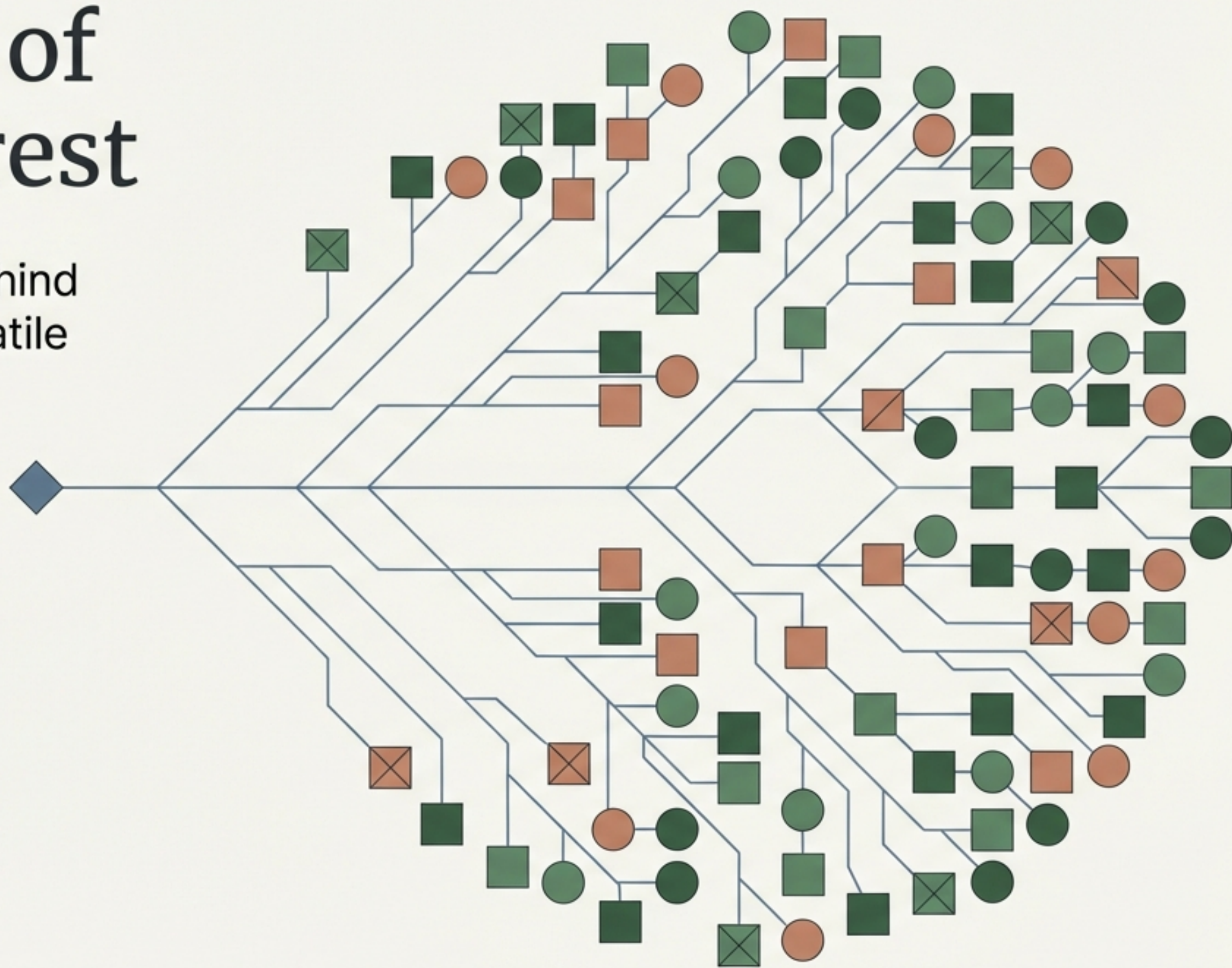


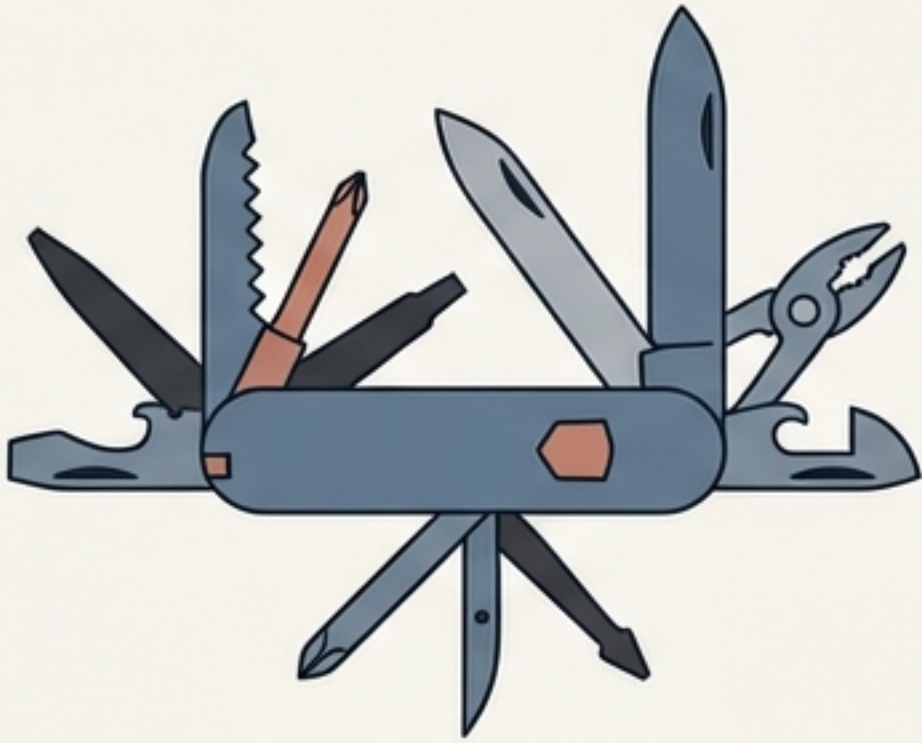
The Anatomy of a Random Forest

Understanding the Intuition Behind Machine Learning's Most Versatile Algorithm



The universal baseline model

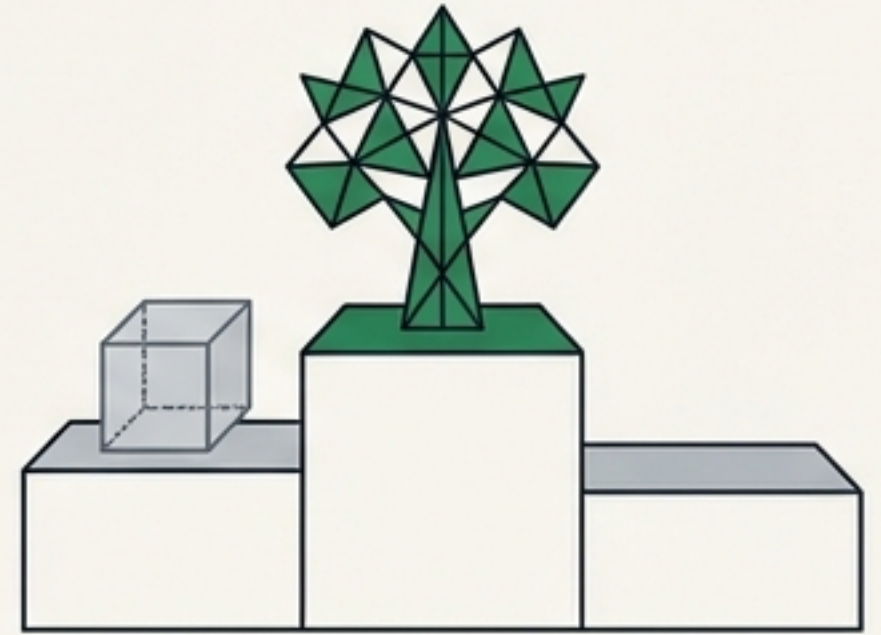
Random Forest consistently delivers robust results across diverse machine learning problems with minimal intervention.



Highly Flexible: Handles both Classification and Regression tasks seamlessly.



Out-of-the-Box Performance: The creator famously noted it requires virtually no hyperparameter tuning to achieve exceptional results.



The Go-To Starting Point: Always a reliable first algorithm to test against any new dataset.

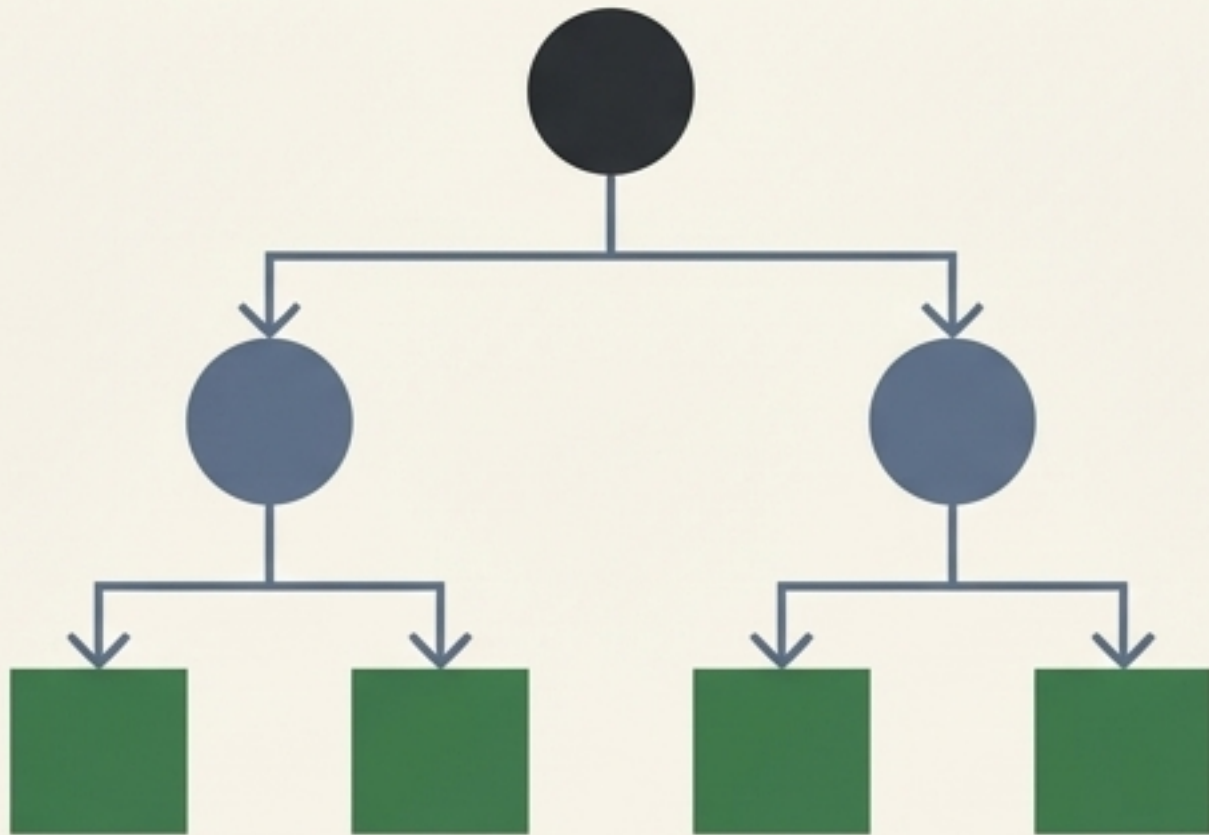
The two essential building blocks

Before understanding the forest, we must understand its roots.

The Random Forest algorithm is a specialised synthesis of two core concepts.

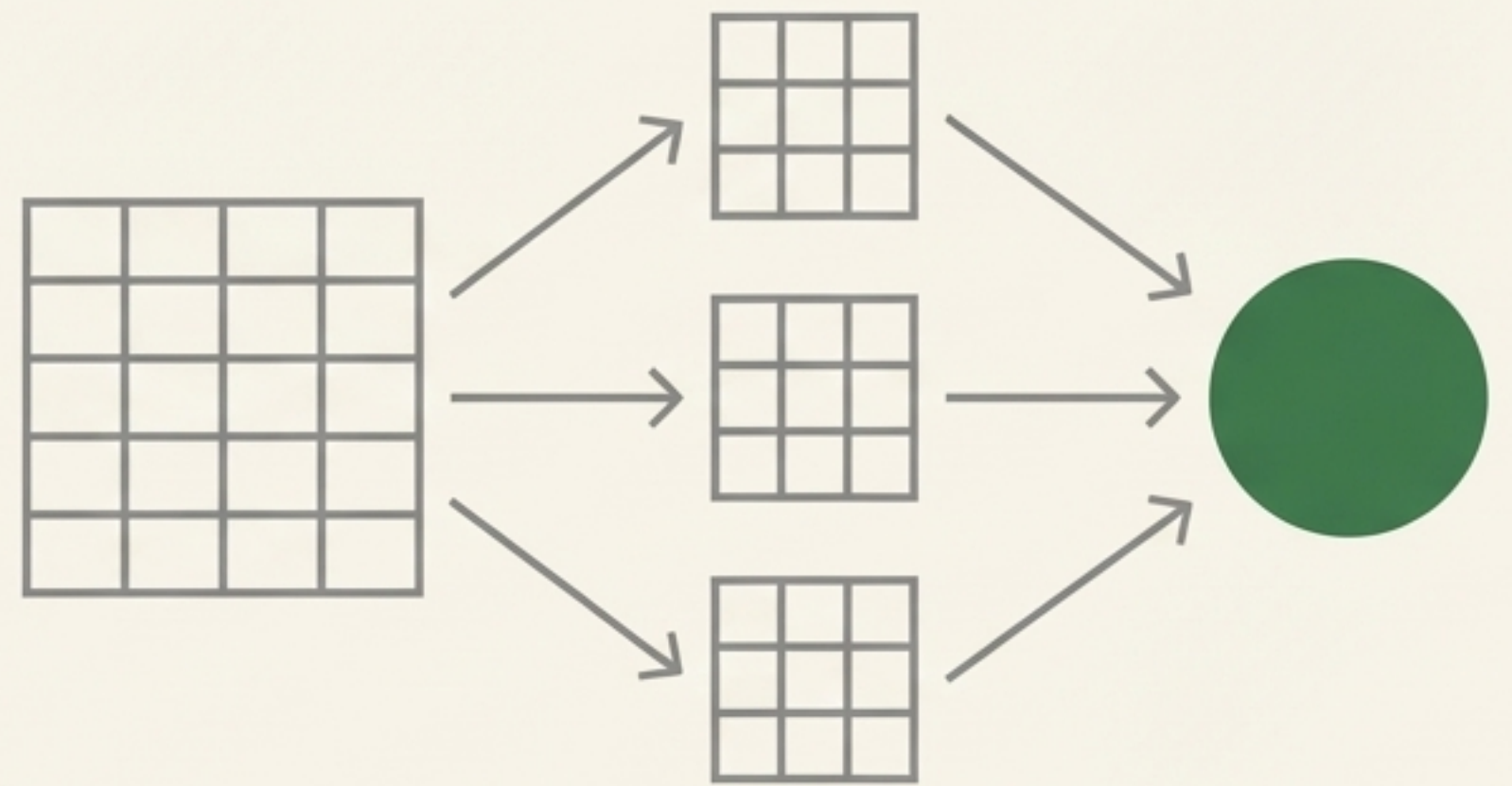
1. Decision Trees

The base model. A flowchart-like structure where data is continuously split according to a certain parameter.



2. Bagging (Bootstrap Aggregation)

An ensemble technique that creates multiple versions of a predictor to build a final, stronger model.

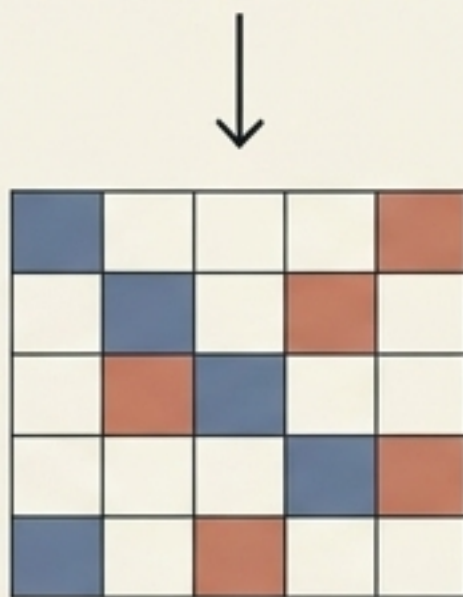


Deconstructing the terminology

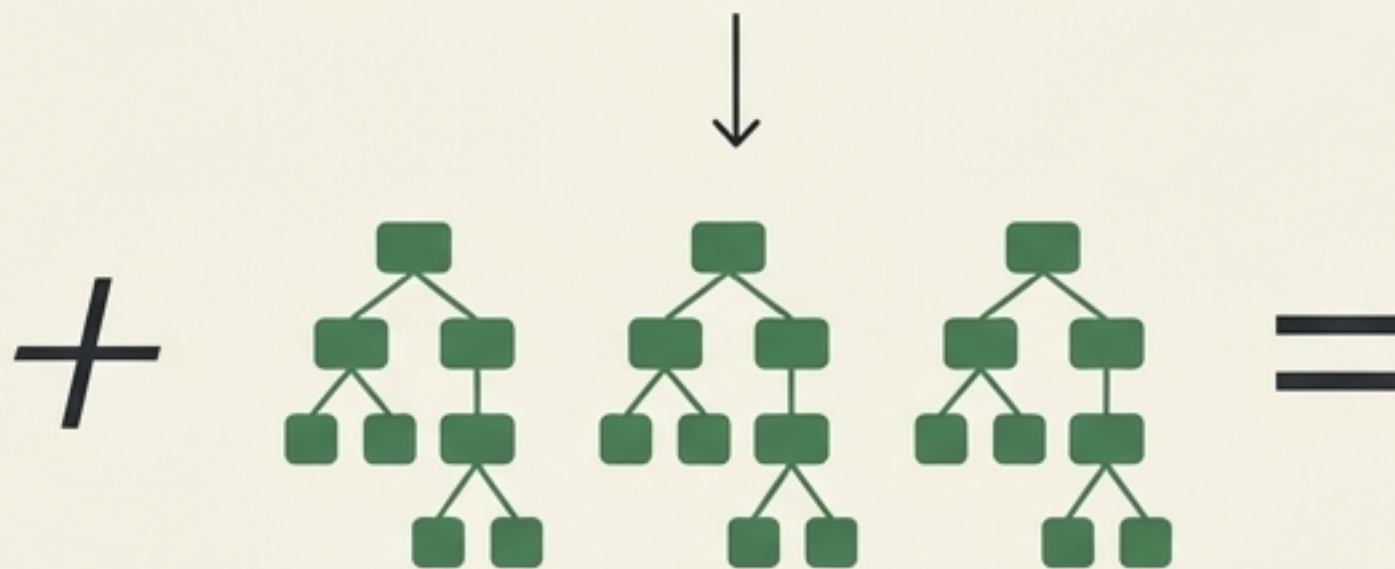
The name "Random Forest" is entirely literal. It perfectly describes the internal architecture of the algorithm.

The Specialisation: When you apply the Bagging technique and exclusively use Decision Trees as your base models, you create a Random Forest.

RANDOM + FOREST



Refers to the deliberate, random selection of data samples (Bootstrapping) fed into the models.

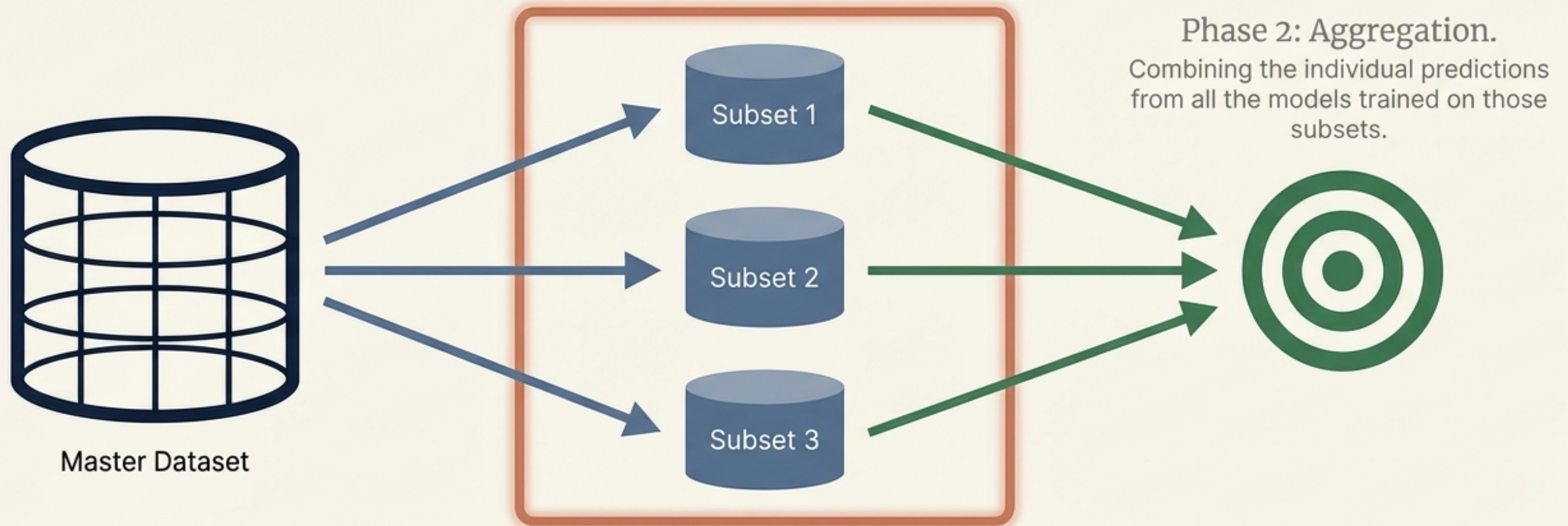


Refers to the grouping of multiple Decision Trees working in parallel.

The core engine: Bootstrap Aggregation

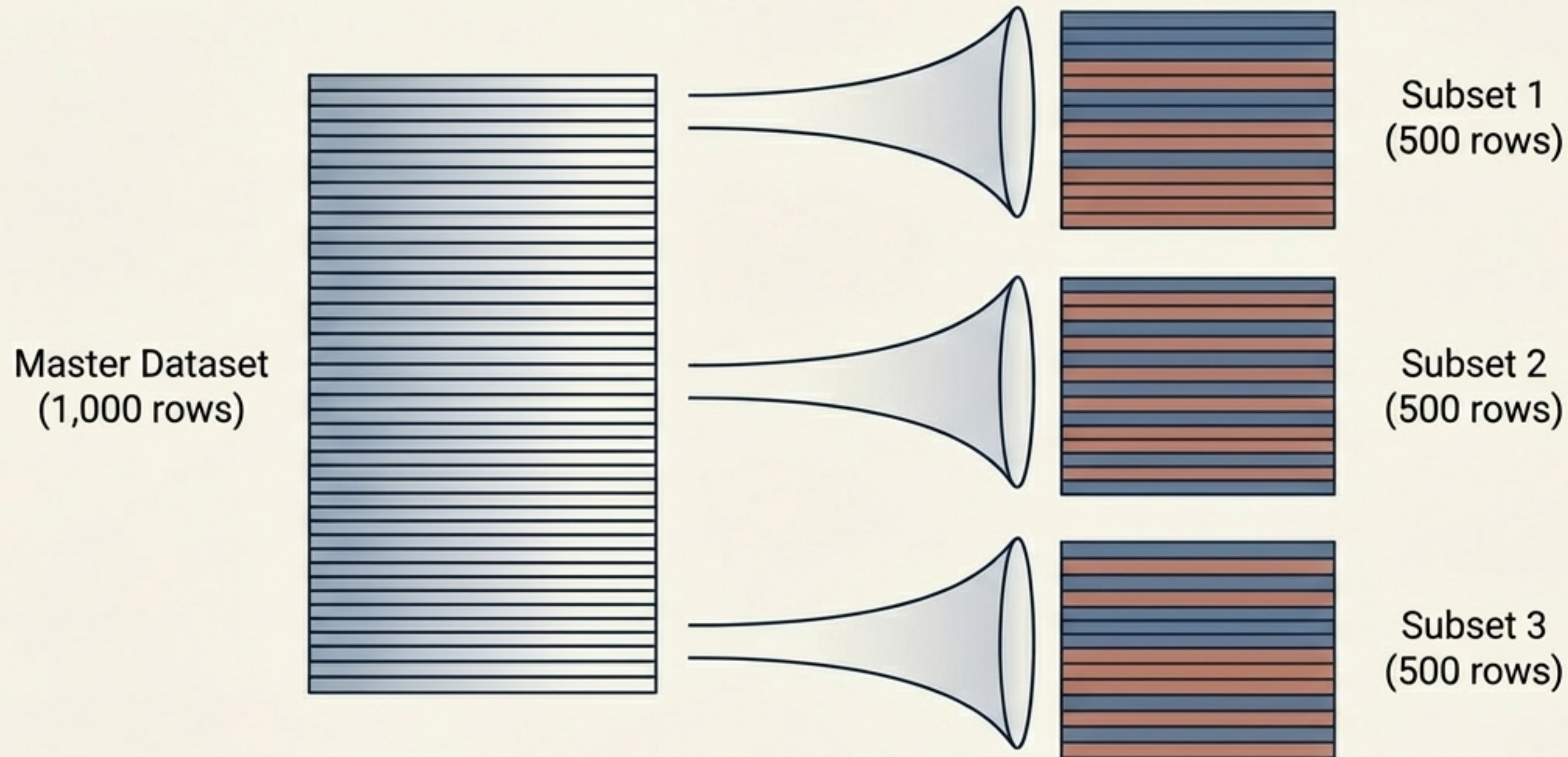
To understand how the forest grows, we must follow the data.
The process is divided into two distinct phases:

Phase 1: Bootstrapping.
Creating unique, smaller subsets of data
from the original master dataset.



Step 1: The Bootstrapping concept

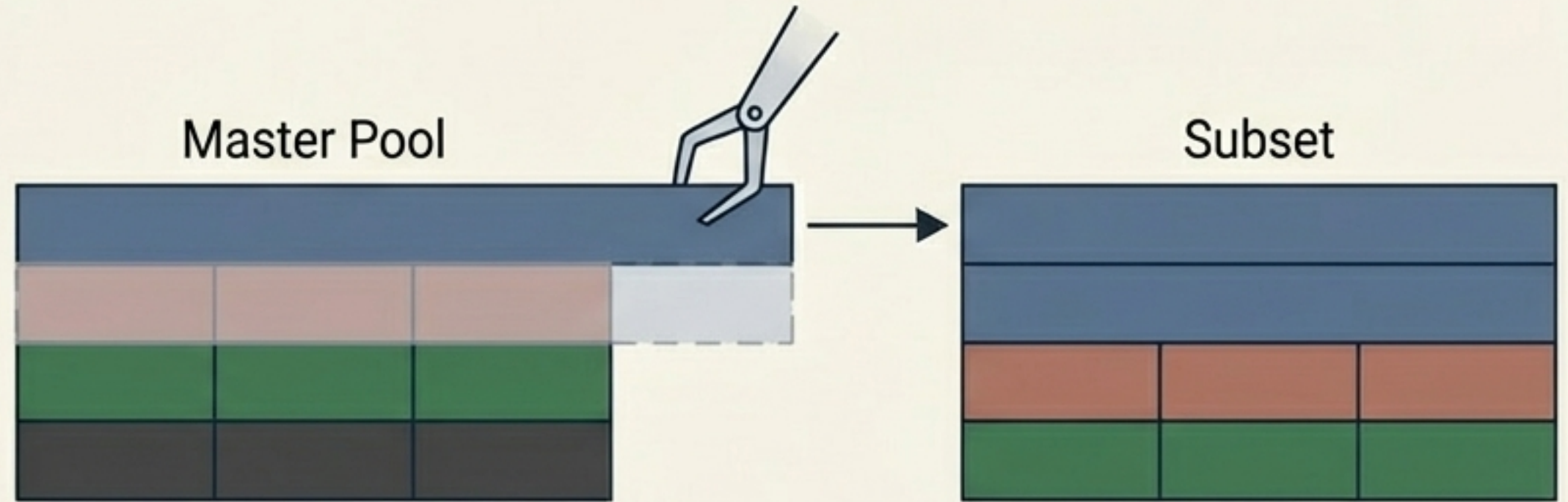
Imagine a master dataset containing 1,000 rows. Instead of feeding the entire dataset to every Decision Tree, we create unique samples. By limiting the data (e.g., providing only 500 rows to a single tree), we ensure that no two trees see the exact same information.



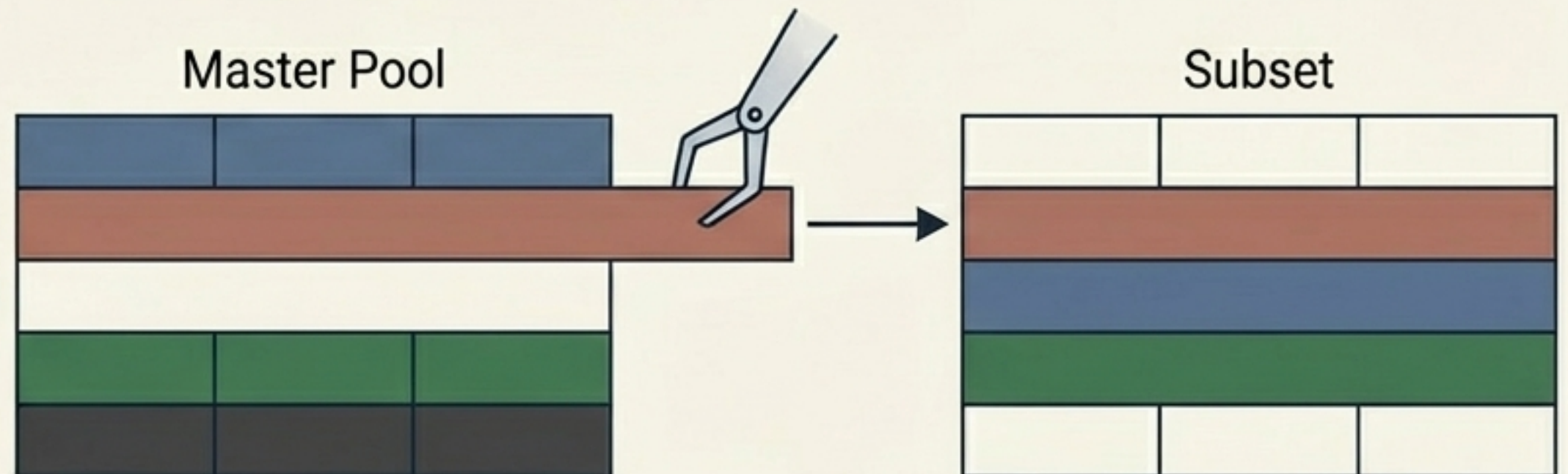
Injecting randomness: Row Sampling

We randomly select rows from the master dataset to build our subsets.
This is done in two ways:

With Replacement: A selected row is returned to the master pool and can be drawn again. The resulting subset may contain duplicate rows.



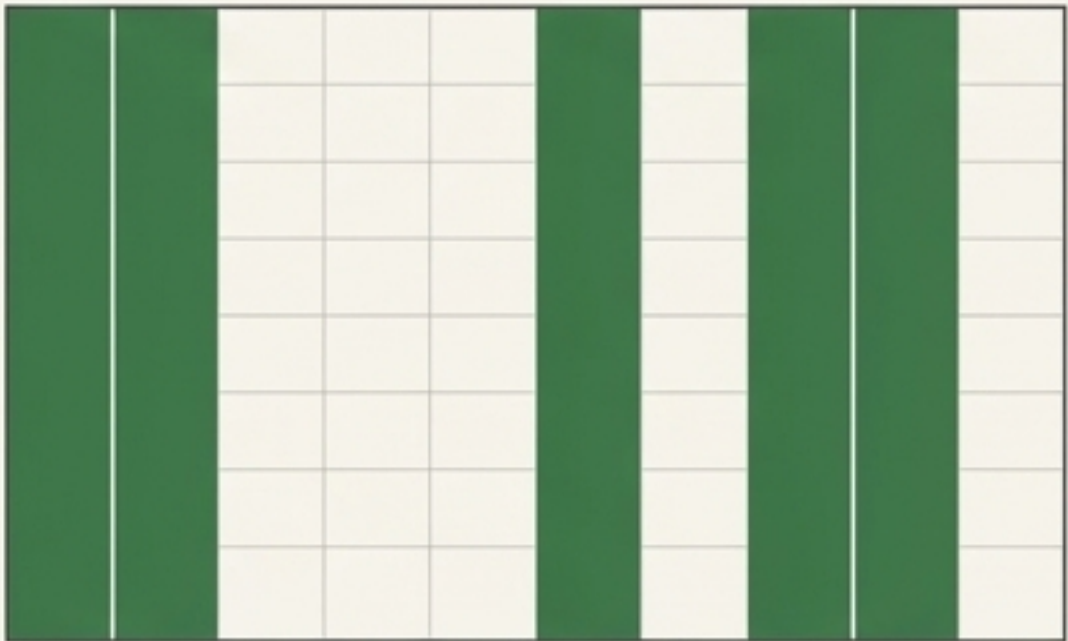
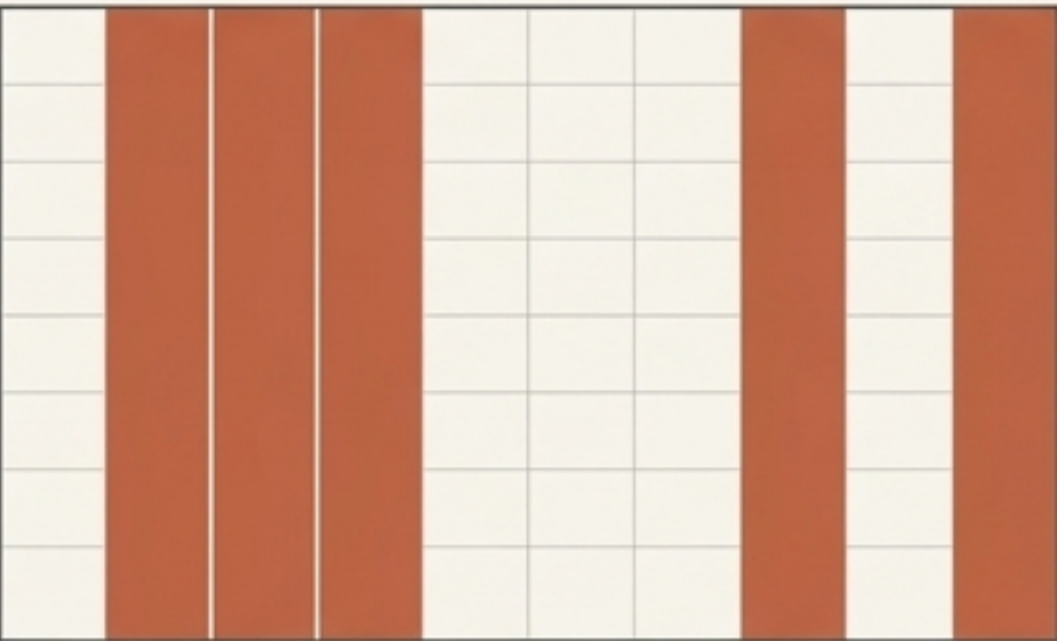
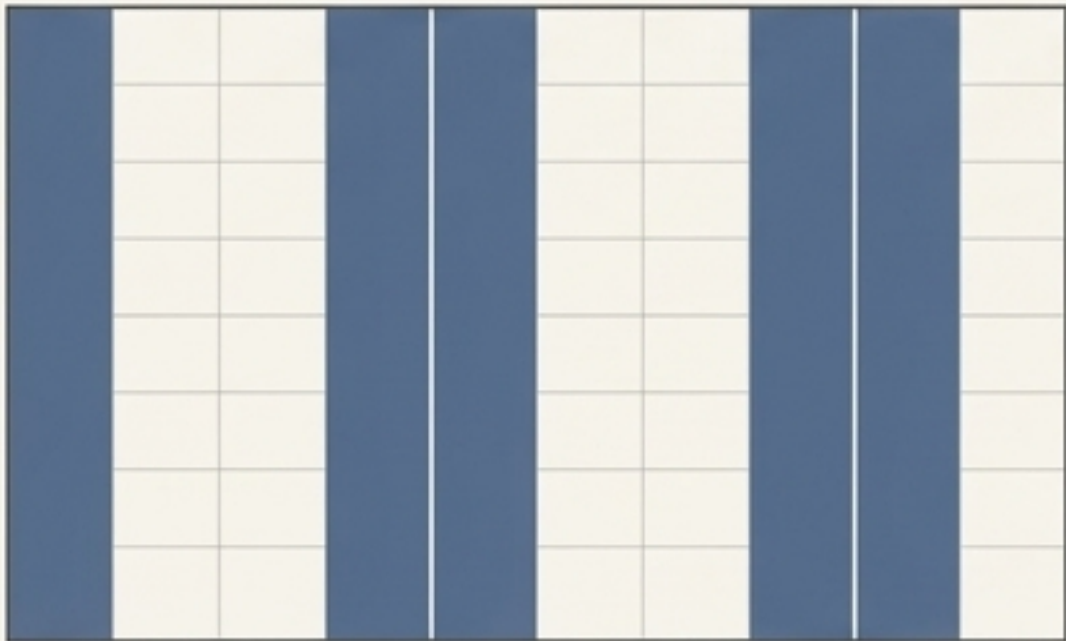
Without Replacement:
A selected row is removed from the pool. Every row in the resulting subset is entirely unique.



Injecting randomness: Feature Sampling

Beyond sampling rows, we can also sample the columns (the features). If our dataset has 10 distinct columns, we might configure the algorithm to only provide 5 random columns to each Decision Tree. This forces the trees to find patterns using entirely different variables.

1	2	3	4	5	6	7	8	9	10

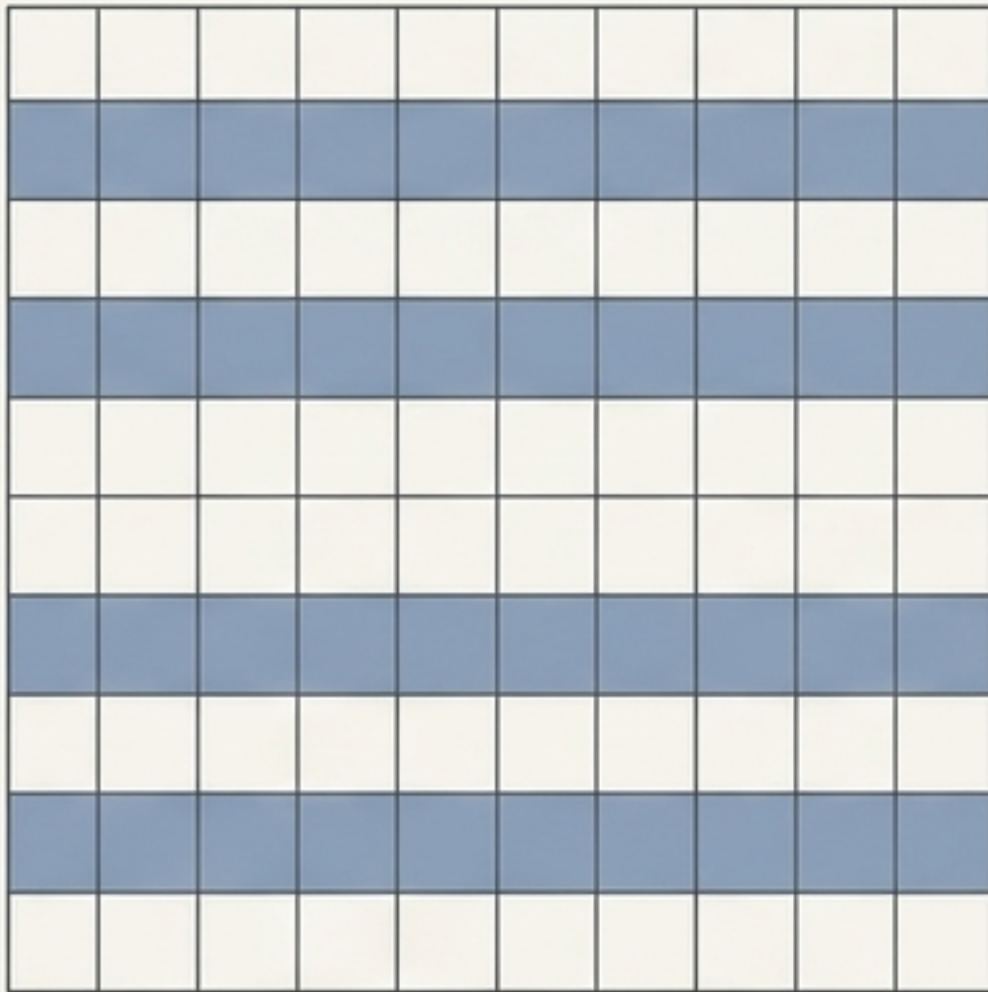


Injecting randomness: Combined Sampling

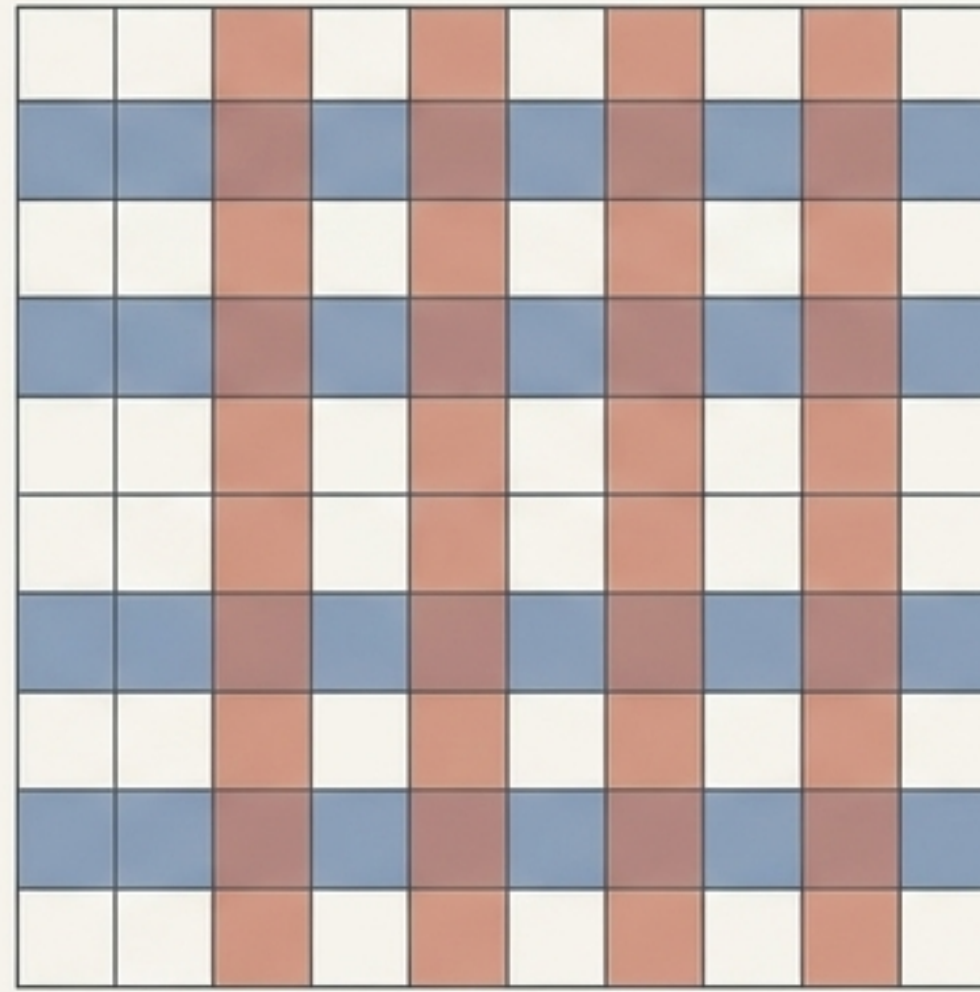
The most comprehensive approach combines both techniques.

We extract a subset of random rows AND a subset of random columns.

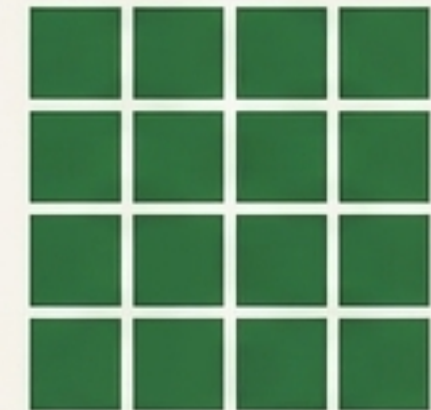
This rigorous sampling ensures maximum variance between the individual Decision Trees, preventing the overall model from simply memorising the original data.



1. Random Row Selection



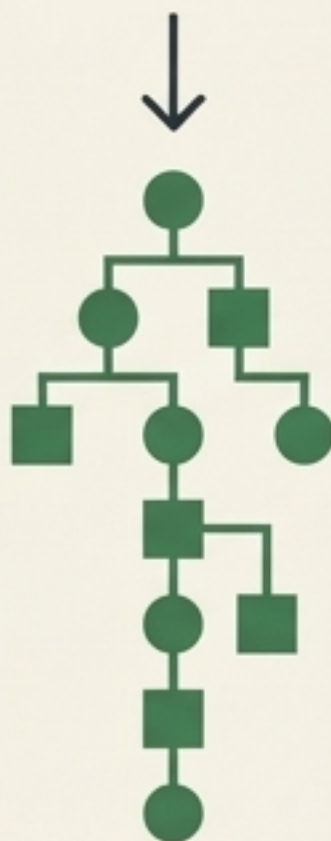
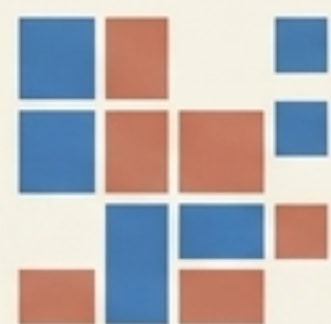
2. Random Column Selection



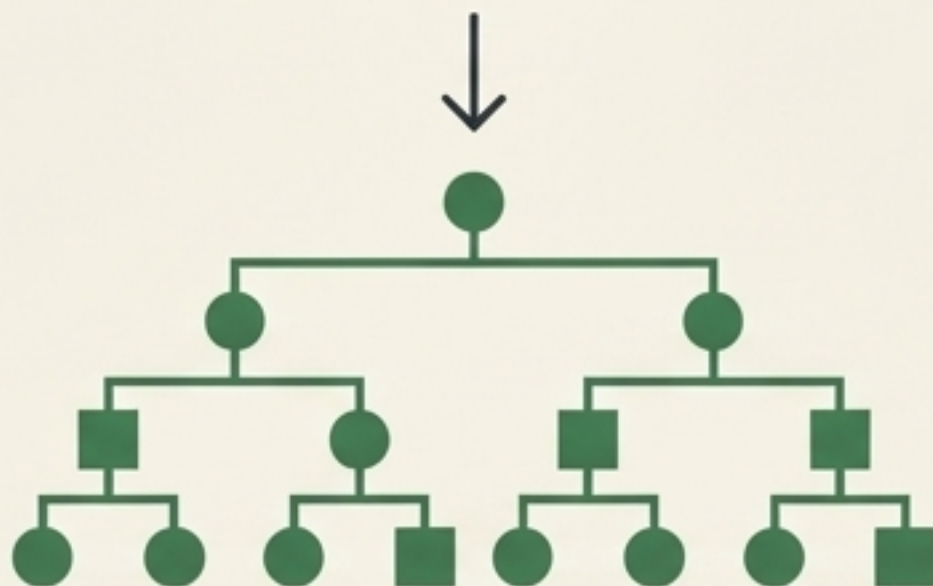
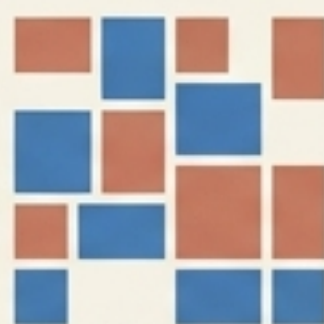
3. Final Sub-sample

Step 2: Training independent models

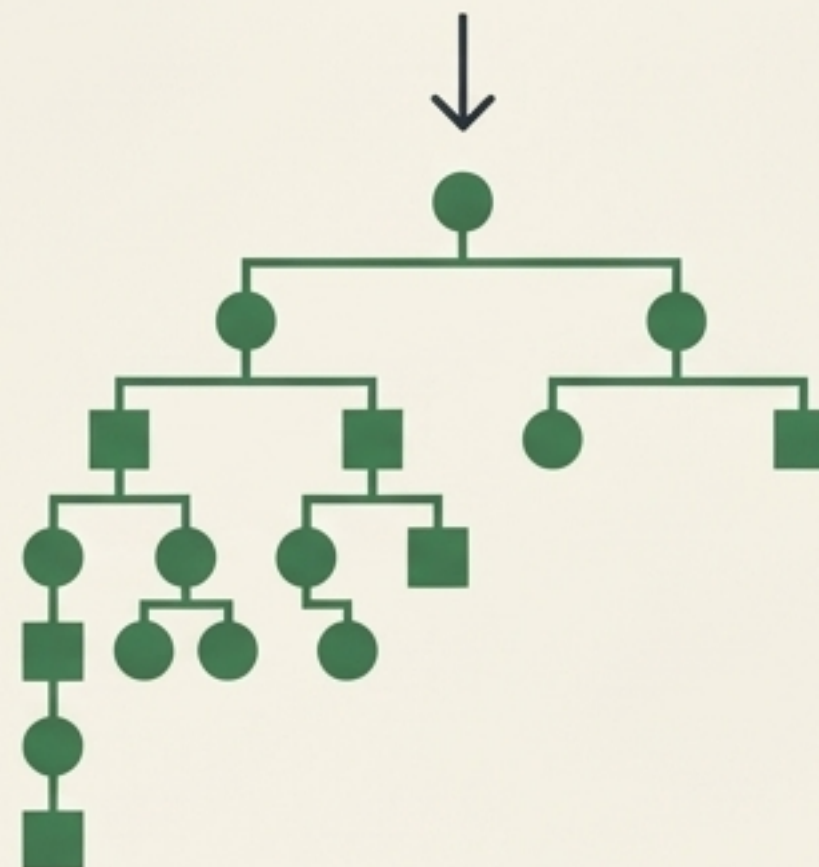
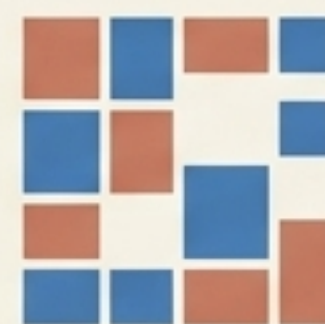
Every generated subset is now fed into its own Decision Tree. Because Subset A looks completely different from Subset B, the resulting trees will grow entirely different branch structures and learn unique internal rules. We now have a truly diverse forest.



Deep and narrow



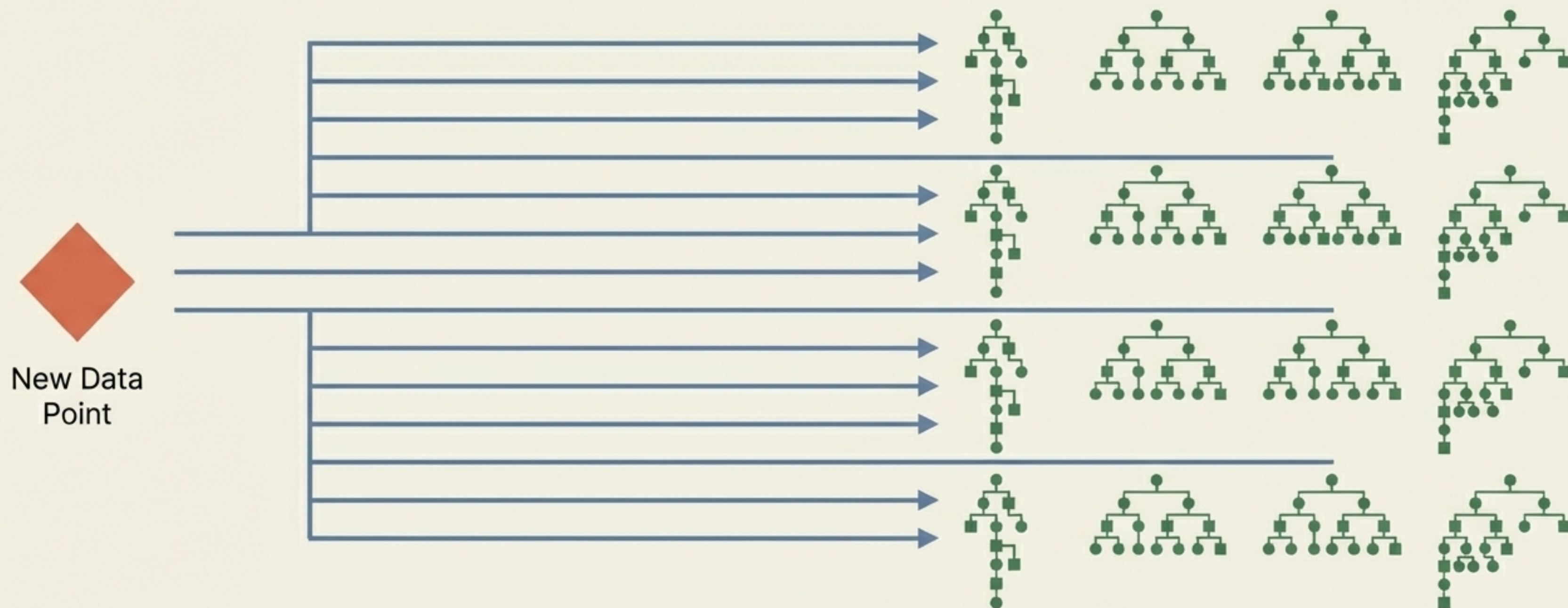
Shallow and wide



Asymmetrical

Step 3: Evaluating a query point

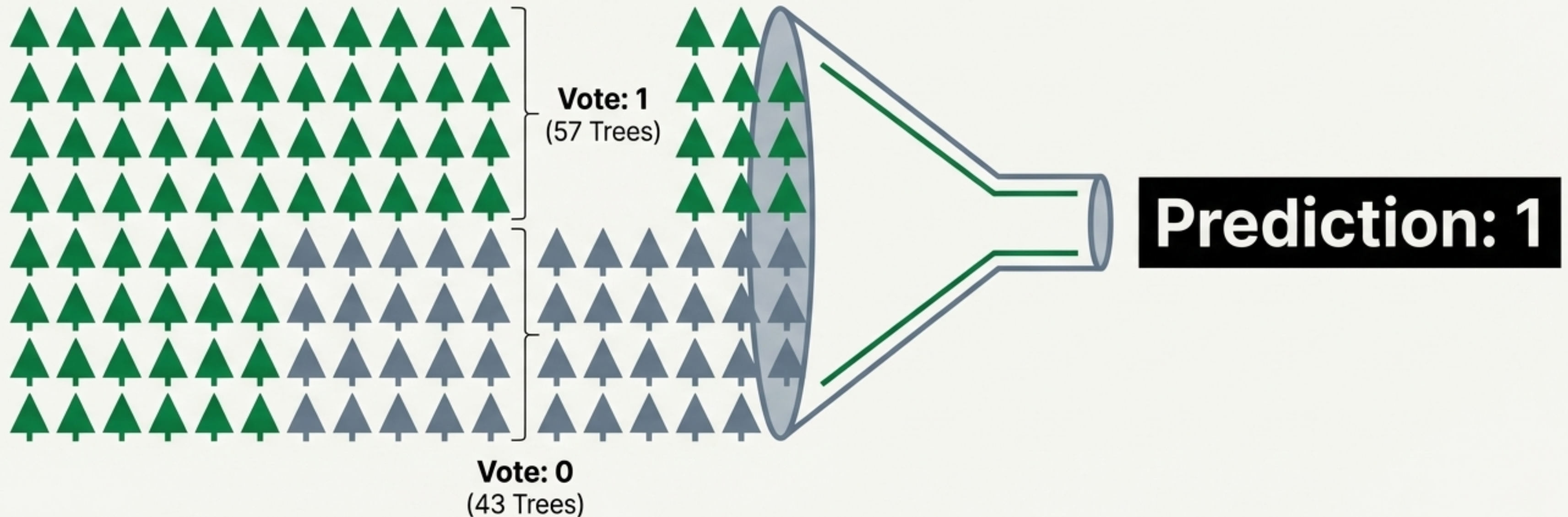
Once the 100+ trees in our forest are fully trained, the model is ready. When a brand new data point (a query point) enters the system, it doesn't just go to one model. It is passed simultaneously through every single Decision Tree in the forest.



Aggregation via majority voting

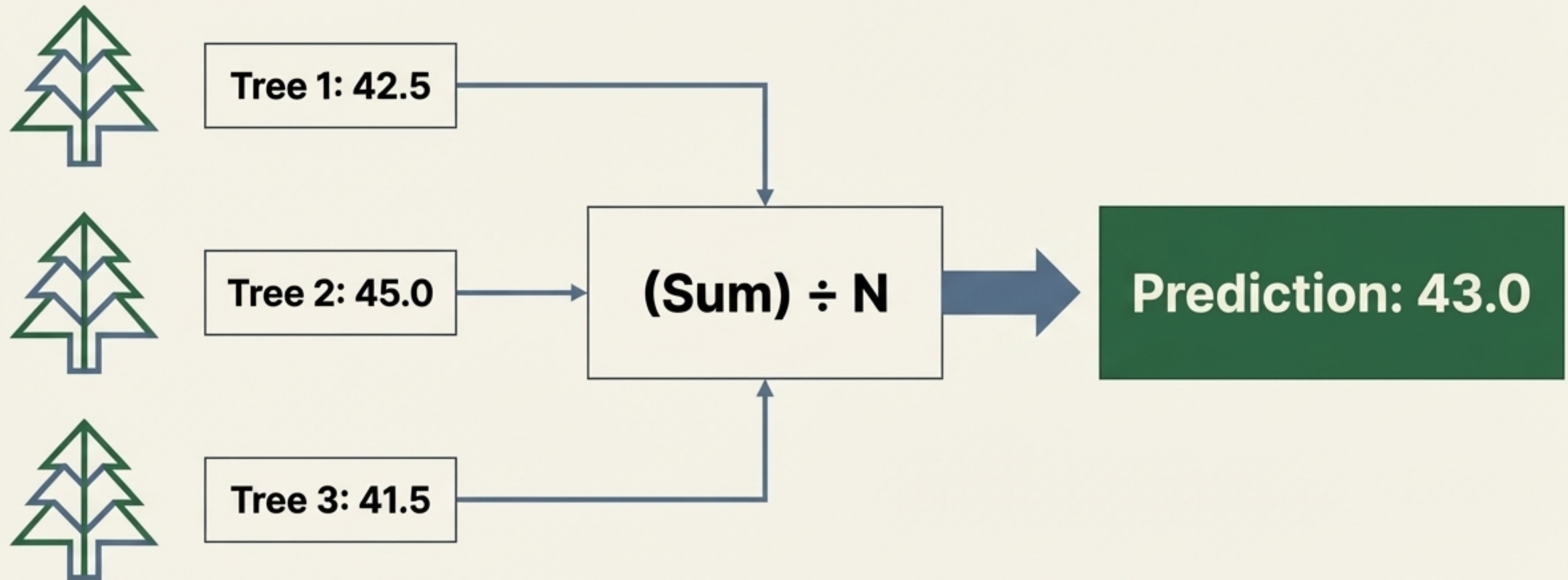
For Classification problems (predicting a category, like 0 or 1), the forest acts as a democracy. Every tree casts a vote based on its individual assessment. The Random Forest aggregates these votes and outputs the majority decision.

Example: Out of 100 trees, 57 predict "1" and 43 predict "0". The final output is "1".



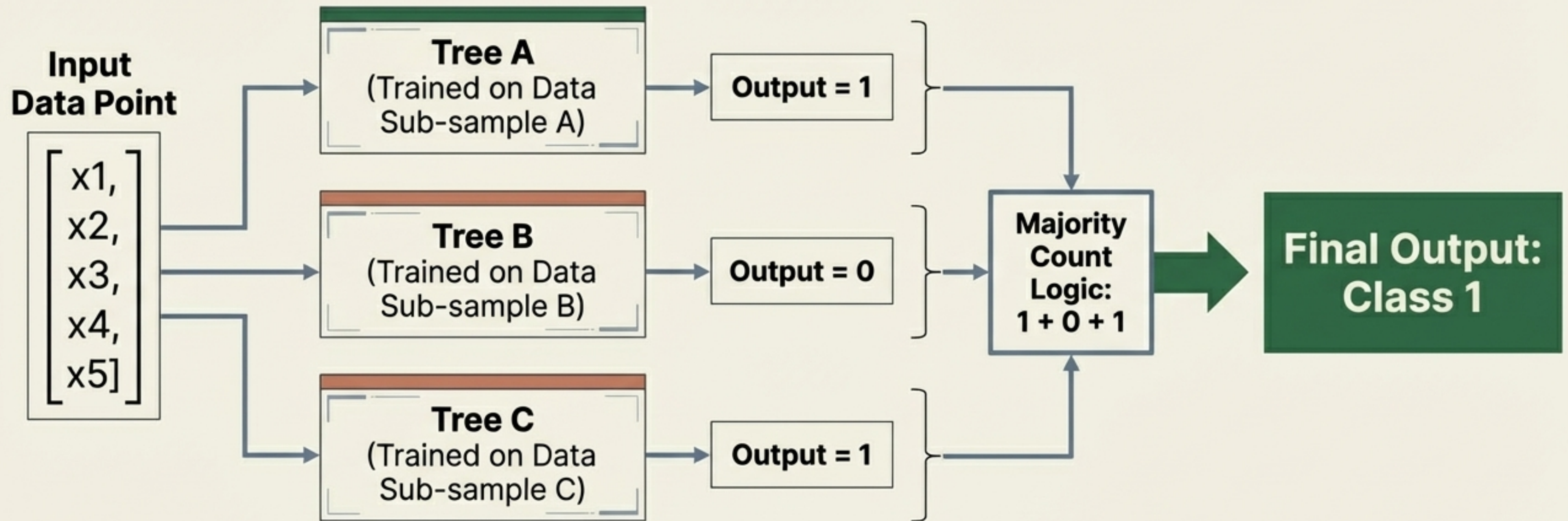
Aggregation via averaging

For Regression problems (predicting a continuous number, like price or temperature), the forest relies on the wisdom of the crowd. Every tree predicts a specific numerical value. **The Random Forest aggregates these numbers** by calculating the mean (average) across all predictions to deliver a final, highly accurate estimate.



Case study: Reaching a consensus

In a controlled test, a master dataset with 5 input features and a target classification column was split using combined sampling. Three completely distinct Decision Trees were trained. When a new query point was introduced, they reached a consensus despite their structural differences.



Strength through diversity

The intuition behind the Random Forest is fundamentally simple: a large collection of diverse, uncorrelated models will always outperform an individual model. By injecting controlled randomness into the data sampling process, the algorithm neutralises the **individual weaknesses of solitary Decision Trees**, resulting in an incredibly robust and highly accurate predictive engine.

